

Ernest van Wijland

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Education

- **PhD Computer Science** **IRIF, Université Paris-Cité**
Paris, France *2024 – Present*
 - PhD student under Claire Mathieu (CNRS) and Vincent Cohen-Addad's (Google Research) supervision.
- **MSc Computer Science, BSc Mathematics** **École Normale Supérieure**
Paris, France *2020 – 2024*
 - Graduated both the MSc (MPRI) and the BSc (double major in Mathematics and Computer Science) with highest honors.
- **Preparatory Class** **Lycée Louis-le-Grand**
Paris, France *2018 – 2020*

Publications

- **Efficient Prover–Estimator Debate via Signed Level-Set Optimization** **In Submission**
Springer, Van Wijland *2026*
- **TCS-Bench: Benchmarking State-of-the-Art Generative AI TCS Research Ability** **In Submission**
*Cohen-Addad *, Paparas *, Van Wijland *, Springer, Canitrot-Paradis, Lin, Woodruff, Kumarappan, Jayaram, Das, Jain, Svensson, Lattanzi, Balunovic, Weber, Mirrokni (*: co-first authors)* *2026*
- **Beyond Worst-Case Online Multi-Level Aggregation** **In Submission**
Mathieu, Naor, Van Wijland *2026*
- **Spectral Dual Fitting for k -Means** **SODA**
Anand, Charikar, Cohen-Addad, Gao, Grandoni, Lee, Sharma, Van Wijland *2027*
- **A $(4 + \varepsilon)$ -Approximation for Euclidean k -Means via Non-Monotone Dual Fitting** **STOC**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2026*
- **An Improved Greedy Approximation for (Metric) k -Means** **FOCS**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2025*
- **Faster Approximation Scheme for Euclidean k -TSP** **SOCG**
Van Wijland, Zhou *2024*
- **Maximizing Nash Social Welfare in 2-Value Instances: Delineating Tractability** **MOOR Journal**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2023*
- **Maximizing Nash Social Welfare in 2-Value Instances** **AAAI**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2022*

Experience

- **Student Researcher** **Google Research**
Paris *2026*

Worked for six months under Silvio Lattanzi and Vincent Cohen-Addad's supervision at Google Research on AI's mathematical abilities.

- PhD Student: Clustering and Network Design**

Paris, France

Currently working under Claire Mathieu and Vincent Cohen-Addad's supervision at IRIF on finding approximation algorithms for clustering and network design problems.

IRIF (Université Paris-Cité)

2024 – Present
- Research Intern: Approximating TSP generalizations**

Paris, France

Worked for a year under Claire Mathieu and Hang Zhou's supervision at IRIF on finding fast approximation schemes for generalizations of the TSP, in Euclidean spaces and general metric spaces.

IRIF (Université Paris-Cité)

2023 – 2024
- Research Intern: Euclidean CVRP, k -TSP**

Palaiseau, France

Worked for a semester under Hang Zhou's supervision in Polytechnique on finding faster approximation schemes for the Euclidean CVRP and k -TSP.

LIX (Polytechnique)

2023
- Research Intern: Approximating 2-WECS**

Lausanne, Switzerland

Worked for a semester under Ola Svensson's supervision in the Combinatorial Optimization team at EPFL on finding a better than 2 approximation for 2-WECS.

THL (EPFL)

2022
- Research Intern: Fair Division for additive instances**

Sarrebruck, Germany

Worked for two months under Kurt Mehlhorn's supervision in the Algorithmic Game Theory team of the MPII on the Fair Division problem for additive instances.

AGT (Max Planck Institut)

2021
- Research Intern: Spectrum of heavy-tailed random matrices**

Paris, France

Worked under Laure Dumaz' supervision on a paper by Ben Arous and Guionnet.

DMA (École Normale Supérieure)

2021

Academic achievements

- Best French Developer Contest 5th place finisher**

Paris, France

 - French national contest in Computer Science, with thousands of participants from every major IT company
 - awarded the 5000€ Genius prize as a high scoring student

2023
- International Olympiad in Informatics Bronze medalist**

Tsukuba, Japan

2018
- Dutch National Olympiad in Informatics winner**

Utrecht, Netherlands

2018
- Innopolis University Olympiad in Informatics finalist**

Innopolis, Russia

2018
- International Olympiad in Informatics participant**

Tehran, Iran

As part of the top 4 High-School students in the Dutch National Olympiad in Informatics.

2017

Outreach

- Member of the France-IOI association**

France

I am involved in organizing the Castor and Algoréa contests, in which more than 700 000 students from elementary school to high-school participate every year. I am also involved in selecting and training the best students that participate in the Informatics Olympiad.

2018 – Present
- Member of the Netherlands-IO association**

Netherlands

2018 – Present

I am involved with the computer science outreach campaign towards Dutch high-school girls and preparing the Dutch girls team for the European Girls Olympiad in Informatics.

Lecturer at the Moroccan Olympiad in Informatics training camp

○ *Morocco*

2019

Gave algorithms lectures to Moroccan middle-school and high-school students.

Other interests

- Active Judo black belt and Tennis player.
- Practicing piano for the past 20 years, part of the chamber music club in the ENS. Founder of the Early Music section.
- Graduated from the Juvisy Music Conservatory in Music Theory, Piano and Ensemble Playing (2018).
- Graduated ENS with a Musicology Minor.